

DOCUMENT INGESTION AGENT

Evaluation Metrics & Rationale

Mortgage Underwriting System

India-focused • Decision-support • Human-in-the-loop

Purpose: document what each Document Ingestion Agent metric measures, why it was selected, and how it supports safe, explainable underwriting evaluation.

1. Evaluation Context

The Document Ingestion Agent is evaluated as a hybrid system. Its spatial extraction core performs multi-format document intake, layout text parsing, structured Pydantic schema extraction, field provenance tracking, deterministic validation, cross-document reconciliation, and finalization with hard safety gates. The deterministic layer remains authoritative for all regulatory and financial rules.

For each application, the key outputs are the classified document type, extracted borrower & property fields, a 0–1 confidence score representing evidence reliability, cross-document contradiction flags, and the safety gate that was triggered, if any.

Output	Range / Type	Meaning
<code>document_type</code>	Enum (8 classes)	Classified document type (PAN, Aadhaar, Payslip, Form 16, ITR, Bank, Property, Unknown)
<code>field_extractions</code>	Pydantic Models	Structured borrower KYC, income, banking aggregates, and property specifications
<code>provenance</code>	Doc + Page + Snippet	Audit trail establishing exact source evidence for every extracted value
<code>confidence</code>	0–1	Evidence reliability and gateway for auto-processing (threshold ≥ 0.85)
<code>gate_triggered</code>	e.g. <code>missing_docs</code>	Safety gate that suspended the case for Human-in-the-Loop review

Two metric families are therefore required: (1) correctness metrics, which compare predictions against labeled ground truth, and (2) operational metrics, which measure how the system behaves from its audit trail without requiring labels.

2. Dataset and Ground Truth

The shipped evaluation dataset contains 40 labeled synthetic cases comprising over 160 individual files. Labels are assigned independently from model predictions according to underwriting design intent, avoiding self-confirmation.

- **Class distribution:** 15 Clean Prime Salaried, 8 Name Discrepancy, 7 Income Variance, 6 Missing Docs, and 4 Low-Quality Scans.
- Borderline cases are deliberately included, including minor spelling variants and variable payslip bonuses.
- Safety-gate cases include `missing_mandatory_docs`, `salary_discrepancy`, `name_contradiction`, and `unmasked_aadhaar`.
- For production evaluation, the synthetic labels should be supplemented with historical underwritten outcomes.

3. Classification Metrics

Classification metrics evaluate whether the predicted document type matches the labeled ground truth across all 8 classes.

3.1 Accuracy and Balanced Accuracy

Accuracy = $\text{correct predictions} \div \text{total cases}$.

Balanced accuracy = **mean of the recall of the classes**.

Significance: Accuracy provides a quick overall correctness measure, while balanced accuracy prevents a dominant class (such as multiple payslips) from hiding poor performance in minority classes (such as Form 16 or Property Deeds).

Why chosen: Underwriting packages contain unequal class distributions. A system that overuses `SALARY_SLIP` could appear accurate while failing on `PROPERTY_DEED`. Balanced accuracy gives each class equal importance.

3.2 Per-Class Precision, Recall and F1

Precision = $TP \div (TP + FP)$. **Recall** = $TP \div (TP + FN)$. **F1** is the harmonic mean of precision and recall.

Class	Precision tells us	Recall tells us
PAN_CARD	How cleanly PANs are identified; limits false KYC.	Whether mandatory tax identity proof is captured.
AADHAAR	How reliably resident Aadhaar cards are identified.	Whether mandatory residential address proof is captured.
SALARY_SLIP	Limits false income proofs; avoids unstructured noise.	Whether all submitted monthly earnings periods are caught.
FORM_16	How justified annual tax certificate extractions are.	Whether statutory annual income proofs are caught.
BANK_STMT	How accurately multi-page transaction ledgers are parsed.	Whether banking cash flows are caught; most safety-critical recall.
PROPERTY	How cleanly collateral title deeds are identified.	Whether property collateral evidence avoids omission.

Why chosen: the document categories do not have equal business consequences. A false classification on a Bank Statement can cause unverified debt calculations, so BANK_STATEMENT recall is a key safety guardrail. F1 provides a compact per-class balance.

3.3 Macro and Weighted Averages

Macro averages treat all classes equally; **weighted averages** weight each class by its support.

Why chosen: macro provides a class-fair view, while weighted averages reflect the actual case mix.

3.4 Confusion Matrix

The confusion matrix tabulates actual versus predicted document types across all 8 classes.

Why chosen: it reveals the type of error rather than reducing performance to one scalar. For example, misclassifying a Bank Statement as a Salary Slip is materially more dangerous than routing an ambiguous scan to UNKNOWN.

3.5 Cost-Weighted Error

The evaluation uses the following asymmetric prototype cost matrix:

Truth \ Pred	PAN	AADHAAR	SALARY	FORM_16	BANK	PROPERTY	UNKNOWN
BANK	8	8	10	6	0	8	2
PAN	0	4	8	6	8	8	2
AADHAAR	4	0	8	6	8	8	2
SALARY	8	8	0	3	10	8	2
PROPERTY	8	8	8	8	8	0	3

Cost-weighted error = $\sum \text{cost}[\text{truth}][\text{prediction}] \div \text{total documents}$.

Why chosen: ordinary accuracy treats every error equally, but underwriting errors are asymmetric. A BANK→SALARY error has the highest prototype cost because it can book unverified credit. Classifying ambiguous files as UNKNOWN carries low cost by providing a safe human-review path.

Important: the cost matrix is configurable and represents an engineering evaluation choice, not an official regulatory cost schedule.

4. Extraction & Validation Metrics

Extraction metrics evaluate whether granular fields and ledgers are extracted verbatim with full source provenance.

4.1 Character Error Rate (CER) and Word Error Rate (WER)

$CER = \text{LevenshteinDistance}(\text{Ref_chars}, \text{Pred_chars}) \div \text{Length}(\text{Ref_chars})$.

Why chosen: isolates whether field parsing failures stem from OCR text degradation versus downstream regex rules.

4.2 Exact Match (EM) and Token F1 Score

Exact Match = 1.0 if normalized string matches ground truth verbatim, else 0.0.

Why chosen: Exact Match is critical for discrete identifiers (PAN: ABCPS1234F, IFSC: HDFC0001234), while Token F1 accommodates minor word permutations in multi-word entities (Employer: 'Tata Consultancy Services Ltd' vs 'Tata Consultancy Services Limited').

4.3 Salary Ledger Arithmetic Reconciliation Rate

Ledger Accuracy = $|\text{Gross Salary} - (\text{Net Salary} + \text{Deductions})| \leq \text{₹}2.00$.

Why chosen: forged payslips frequently exhibit arithmetic mismatches. Deterministic math checks trap fraud before credit scoring.

4.4 Field Provenance Coverage Rate

Provenance Coverage = $\text{Extracted fields with valid (Doc, Page, Snippet)} \div \text{Total extracted fields}$.

Why chosen: ensures 100% auditability for regulatory compliance and rapid human-in-the-loop verification.

5. Cross-Document Reconciliation Metrics

Reconciliation metrics evaluate whether the agent successfully cross-examines data across independent documents.

5.1 Name Resolution and Income Reconciliation Accuracy

Significance: Evaluates fuzzy matching across KYC, salary slips, and bank statements, and reconciles monthly net salary against detected bank deposit credits within $\pm 5\%$ tolerance.

5.2 Contradiction Detection Recall (Safety-Critical Guardrail)

Contradiction Recall = $\text{True Contradictions Detected} \div \text{Total Actual Contradictions}$.

Why chosen: This is the most safety-critical recall metric. Failing to catch a salary discrepancy or identity conflict allows corrupted data to reach downstream credit and decision agents. Target benchmark is $\geq 98.0\%$.

6. Calibration Metrics

Calibration asks whether the confidence score means what it claims. A confidence of 0.90 should correspond approximately to a 90% probability of extraction correctness. This is vital because the pipeline uses confidence ≥ 0.85 as the gateway for auto-processing.

6.1 Brier Score

Brier score = Mean squared error between confidence and correctness (1 for correct, 0 for incorrect). Lower is better; 0 is perfect.

Why chosen: provides one scalar reflecting correctness and confidence quality. Overconfidence is penalized heavily.

6.2 Expected Calibration Error (ECE)

ECE = $\sum (|B_m| \div N) \times |\text{acc}(B_m) - \text{conf}(B_m)|$ across 5 confidence bins. 0 represents perfect calibration.

Why chosen: directly tests the operational meaning of the 0.85 confidence threshold across operational tiers.

6.3 Reliability Bins

Reliability bins report the confidence interval, case count, mean confidence, and actual accuracy for each bin:

Confidence Interval	Case Count	Mean Confidence	Actual Accuracy	Calibration Gap
0.00 – 0.20	0	0.1000	0.0000	0.0000
0.20 – 0.40	0	0.3000	0.0000	0.0000
0.40 – 0.60	0	0.5000	0.0000	0.0000
0.60 – 0.80	12	0.6973	1.0000	0.3027
0.80 – 1.00	28	0.9142	1.0000	0.0858

Why chosen: diagnostic view behind ECE showing calibration across confidence bands.

7. Runtime / Operational Metrics

Operational metrics require no ground-truth labels and measure pipeline health from audit logs:

Metric	Definition / Significance	Why It Was Chosen
stp_rate	Straight-Through Processing: $1 - \text{hitl_rate}$.	Shows efficiency gained through automation.
hitl_escalation_rate	Applications routed to human review ÷ all cases.	Measures human-review load and queue depth.
mean_confidence	Average overall confidence across applications.	Provides overall view of evidence quality.
below_threshold_rate	Share of applications below 0.85 confidence.	Shows how often low-confidence gate fires.
gate_trigger_counts	Count per safety gate trigger (Missing, Mismatch, Variance).	Identifies dominant operational defect.
provenance_coverage	Share of extracted values linked to source bounding text.	Guarantees 100% auditability for compliance.

8. Current Baseline on the Shipped Benchmark Dataset

Evaluation Metric	Baseline Value	Target Benchmark
Document Classification Accuracy	0.9938 (99.38%)	≥ 0.950
Classification Balanced Accuracy	0.9956 (99.56%)	≥ 0.950
Asymmetric Cost-Weighted Error	0.0124	≤ 0.150
Field Extraction Exact Match (EM)	0.9770 (97.70%)	≥ 0.920
Field Extraction Token F1 Score	0.9878 (98.78%)	≥ 0.950
OCR Character Error Rate (CER)	0.0142 (1.42%)	≤ 0.030
Contradiction Detection Recall (Safety)	1.0000 (100.00%)	≥ 0.980
Contradiction Precision	1.0000 (100.00%)	≥ 0.900
Brier Score	0.0346	≤ 0.100
Expected Calibration Error (ECE)	0.1489	≤ 0.150
Straight-Through Processing (STP) Rate	0.3750 (37.50%)	Reflects 40-case mix
HITL Escalation Rate	0.6250 (62.50%)	Traps all 25 edge cases

These values are baseline results for the shipped benchmark dataset and should be interpreted as an engineering benchmark.

9. Why This Metric Set Was Chosen

- **Safety:** Contradiction recall, confusion matrices, and cost-weighted error emphasize the asymmetric harm of corrupted borrower financials.
- **Class fairness:** balanced accuracy and macro metrics prevent payslip-heavy document distributions from hiding weak classification on rare deeds.
- **Extraction fidelity:** Exact Match, Token F1, and CER test whether granular entities and financial figures are extracted without distortion.
- **Trust in automation:** Brier, ECE, and reliability bins test whether confidence is reliable enough to support the 0.85 auto-processing threshold.
- **Operational health:** STP rate, gate triggers, and escalation rates expose document quality problems that static test sets cannot show in real time.
- **Auditability:** Provenance coverage guarantees full source verification rather than treating the LLM/OCR as an unverifiable authority.

10. Future Evaluation Scope

- **Multilingual KYC extraction:** benchmark performance across Hindi, Tamil, Telugu, and Marathi regional identity documents.
- **Digital tampering detection:** quantify precision/recall on image splicing, font inconsistencies, and PDF metadata modifications.
- **Per-segment slicing:** evaluate performance by employer tier, income band, scan resolution, and bank statement format.
- **Latency and cost:** tokens, wall-clock time, and OCR compute cost per multi-page document bundle.

11. Implementation / Reproducibility

The metrics can be reproduced using the project's evaluation CLI or API:

```
python -m src.evaluation.eval_cli
python -m src.evaluation.eval_cli --output evaluation_report.json
python -m pytest tests/ -v
```

12. Scope and Governance Note

The Document Ingestion Agent is an evidentiary extraction and reconciliation component within a human-in-the-loop underwriting architecture. The metric framework is designed to evaluate engineering behavior, extraction fidelity, safety guardrails, calibration, and operational health. The prototype tolerance bands ($\pm 5\%$) and cost matrices are engineering choices; they should not be represented as official RBI rules or regulatory requirements.